

Procedure of U-score Calculation

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I. PROCEDURE FOR U-SCORE

For the U-score calculation, we follow the CEC 2026 constrained optimization evaluation protocol¹; each run stores the best-so-far value Min_EV and the lowest constraint violation LCV after initialization and then every $10D$ evaluations until MaxFES . Therefore, each run contains 2001 saved points. Let $E_{a,r}^{(f)}(t)$ and $C_{a,r}^{(f)}(t)$ denote the saved Min_EV and LCV, respectively, for algorithm a , run r , function f , and sampling point t . The LCV is computed as:

$$C_{a,r}^{(f)}(t) = \min_{1 \leq i \leq NP} CV(P_i(t)), \quad (1)$$

where $P_i(t)$ is the i th individual in the population at sampling point t . If no feasible solution exists at a sampling point, Min_EV is recorded as NaN.

Let m be the number of compared algorithms and n be the number of independent runs. For each function, all mn trials are compared pairwise. The number of trial pairs is

$$N_{\text{pair}} = \frac{mn(mn - 1)}{2}. \quad (2)$$

For two trials x and y , let E_x^* and C_x^* denote the final saved error and constraint violation at MaxFES . A trial is feasible if $C_x^* \leq \epsilon_{\text{CV}}$ and E_x^* is finite.

For pairwise scoring, define the smaller-is-better comparison function

$$\rho(u, v; \epsilon) = \begin{cases} 1, & u < v - \epsilon, \\ 0.5, & |u - v| \leq \epsilon, \\ 0, & u > v + \epsilon. \end{cases} \quad (3)$$

The value $\rho(u, v; \epsilon)$ represents the point awarded to the first trial when comparing u and v .

The pairwise accuracy point awarded to trial x against trial y is computed as

$$A(x, y) = \begin{cases} \rho(E_x^*, E_y^*; \epsilon_{\text{EV}}), & x, y \text{ are feasible,} \\ \rho(C_x^*, C_y^*; \epsilon_{\text{CV}}), & x, y \text{ are infeasible,} \\ 1, & x \text{ is feasible and } y \text{ is infeasible,} \\ 0, & x \text{ is infeasible and } y \text{ is feasible,} \end{cases} \quad (4)$$

Therefore, feasible trials are always preferred over infeasible trials. If both trials are feasible, the final Min_EV values are compared. If both trials are infeasible, the final LCV values are compared.

The accuracy score of algorithm a on function f is the sum of all pairwise accuracy points obtained by its n trials:

$$\mathcal{A}_a^{(f)} = \sum_{x \in \mathcal{T}_a^{(f)}} \sum_{\substack{y \in \mathcal{T}^{(f)} \\ y \neq x}} A(x, y), \quad (5)$$

where $\mathcal{T}_a^{(f)}$ is the set of trials of algorithm a on function f , and $\mathcal{T}^{(f)}$ is the set of all trials from all algorithms on function f .

The speed score is computed using the first sampling point at which a trial reaches a pairwise reference value. For two final feasible trials, the reference value is the worse final error value:

$$\theta_E(x, y) = \max(E_x^*, E_y^*). \quad (6)$$

The first reaching index of trial x is then defined as

$$\tau_E(x, y) = \min \{t \mid E_x(t) \leq \theta_E(x, y) + \epsilon_{\text{EV}}\}. \quad (7)$$

If the reference value is never reached, $\tau_E(x, y)$ is set to infinity.

If one or both trials are infeasible at the final sampling point, the comparison is based on the lowest constraint violation. The pairwise reference value is

$$\theta_C(x, y) = \max(C_x^*, C_y^*), \quad (8)$$

and the first reaching index is

$$\tau_C(x, y) = \min \{t \mid C_x(t) \leq \theta_C(x, y) + \epsilon_{\text{CV}}\}. \quad (9)$$

The speed point awarded to trial x against trial y is therefore

$$S(x, y) = \begin{cases} \rho(\tau_E(x, y), \tau_E(y, x); 0), & x, y \text{ are feasible,} \\ \rho(\tau_C(x, y), \tau_C(y, x); 0), & \text{otherwise.} \end{cases} \quad (10)$$

Thus, the trial that reaches the pairwise reference value earlier receives one point. If both trials reach it at the same sampling point, both trials receive 0.5 points.

The speed score of algorithm a on function f is computed as

$$\mathcal{S}_a^{(f)} = \sum_{x \in \mathcal{T}_a^{(f)}} \sum_{\substack{y \in \mathcal{T}^{(f)} \\ y \neq x}} S(x, y). \quad (11)$$

The U-score of algorithm a on function f is the sum of its accuracy and speed scores:

$$\mathcal{U}_a^{(f)} = \mathcal{A}_a^{(f)} + \mathcal{S}_a^{(f)}. \quad (12)$$

The final U-score over all $K = 28$ functions is calculated as

$$\mathcal{U}_a = \sum_{f=1}^K \mathcal{U}_a^{(f)}. \quad (13)$$

¹<https://github.com/P-N-Suganthan/2026-CEC>

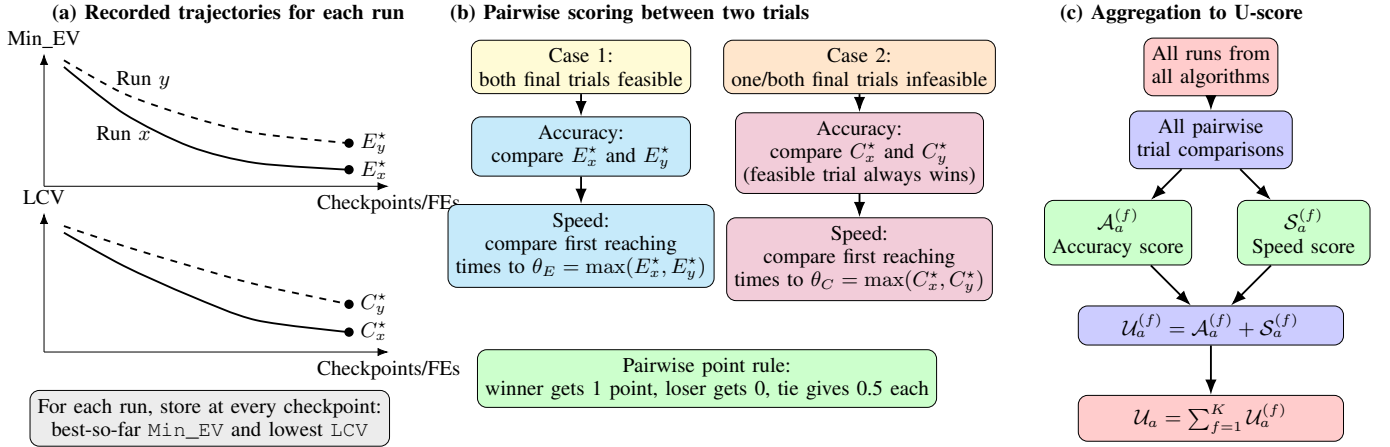


Fig. 1. Illustration of the pairwise U-score procedure for CSOPs. (a) For each run, the best-so-far feasible objective value Min_EV and the lowest constraint violation LCV are recorded at fixed checkpoints. (b) For a pair of trials, the accuracy and speed components are computed in a feasibility-aware manner. If both final trials are feasible, accuracy compares the final Min_EV values and speed compares the first checkpoint at which each trial reaches the pairwise reference level $\theta_E = \max(E_x^*, E_y^*)$. If one or both final trials are infeasible, accuracy is based on the final LCV values (with a feasible trial always preferred), and speed compares the first checkpoint at which each trial reaches $\theta_C = \max(C_x^*, C_y^*)$. (c) Pairwise accuracy and speed points are accumulated over all trial pairs to obtain the function-wise score $\mathcal{U}_a^{(f)}$, which is then summed over all functions to produce the final U-score.

For each function, algorithms are ranked according to $\mathcal{U}_a^{(f)}$ in descending order. Rank 1 denotes the best algorithm. The total rank is obtained by summing the function-wise ranks across all benchmark functions. The procedure of U-score calculation is illustrated in Fig. 1.